Working with AI Agents

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AIs are changing rapidly, but the default approach is similar

1. **Just fool around with it:** you don't need a tutorial.
2. **Everything is meta:** if you don't know how to use it, ask it.
3. **Aim for persistence:** leave behind instructions and project memory.
4. **Teach it to work more like you.**
5. **Plan, iterate, and show it examples of your work.**

Six months later: I would tweak some of the above slightly, but they are all still valid.

Your AI use is a signal for other people

If you send me unfinished AI work, I learn something about you.

- Be a useful middleman: if someone needs generic AI output, they can produce it themselves.
- **What is the researcher's added value?**
- The standard of our work has to go up (brutal but true).
- Be transparent about your contributions and AI's contributions.

When is it okay to let AI write code?

Discuss!

- When you have **100% verification**?
- When you **understand the method**?
- When you **don't care about implementation**?

Basic workflow with agents

Terminal → project directory → agent → code / `.md` files / shell

- Integration across tools is getting better.
- Learn to connect sources of context *(I like Obsidian for this)*.
- Large premium to experimenting — and experimentation is easy.
- For simple work, the limiting factor is often your ability to supervise.

My mess: cloud and terminal agents still do not feel connected

Short demo with BEAR

What should humans focus on?

The more context AI gets, the less valuable humans are as conduits for information.

Human value increasingly lies in taste, judgement and supervision.

- **Taste:** What is actually interesting? What is high value?
- **Judgement:** Which objections matter? Which method is appropriate?
- **Supervision:** What should we do next? Did what we just did make sense?

Make it behave more like you — transparently

- **Short iteration loops:** agent works → I review → feedback → next task.
- Keep loops short enough that you understand what happened.
- Detailed feedback is cheap: **dictation and stream-of-consciousness instructions work well**.
- **Where do instructions live?**
  - Global config
  - Repo-specific instructions (`AGENTS.md`)
  - Skills

What am I doing while the agent works?

We used to do thinking by typing. Now we just need to do thinking.

- What is worth doing next?
- Are we still answering the scientific question?
- What should go in the paper?
- Is this result plausible or interesting?
- Which rabbit holes are worth pursuing?

Thanks

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